ReCycleGuard

Agentic AI for E-Waste EPR Compliance Reconciliation

Agentic AI (26CD7PEAAI) · B.E. CSE (Data Science), VII Semester, BMSCE
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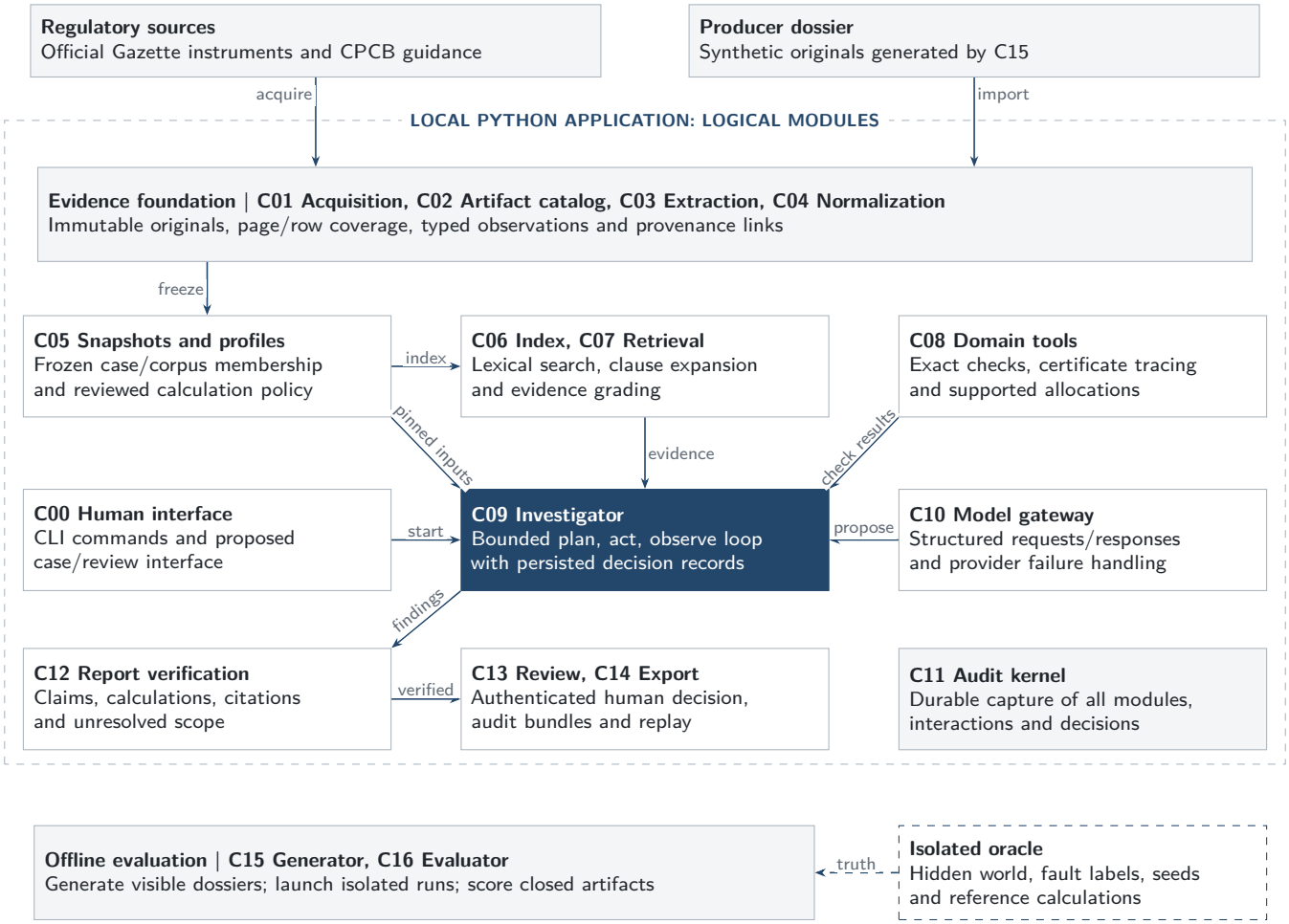
1 Decision, scope and architecture

Decision supported. Which discrepancy should the compliance officer investigate next, and what verification or correction does the evidence support? The output is a traceable working paper with calculations, citations, uncertainty and named next-action owners.

Scope. One synthetic producer; closed FY 2024–25; up to two specific reviewed EEE codes; supplied Au/Cu/Al/Fe obligation buckets and separately dated historical liabilities. Gold is a fixed supplied bucket; gold-specific adjustments are unsupported. The system does not derive statutory targets from sales. A bounded single investigator operates within a human-reviewed workflow.

Excluded. Live portal filing, certificate trading, OTP interaction, autonomous legal certification and compensation calculation. The application, dossiers and experiments are *planned*; no implementation results are claimed.

Component view: local application and evaluation boundary



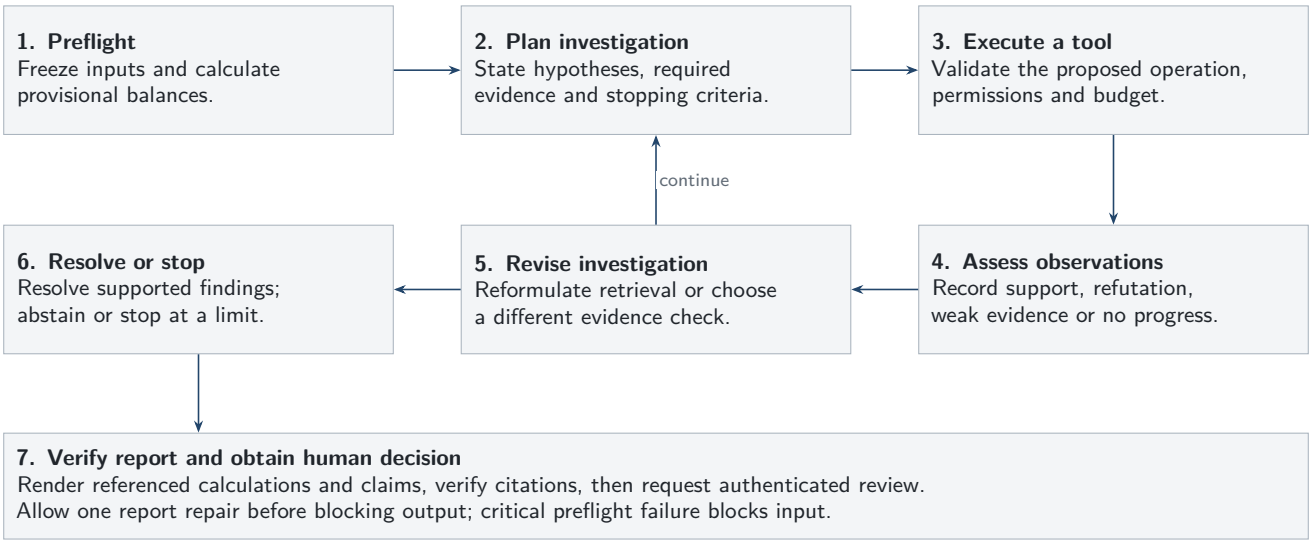
Key. Boxes group logical components; solid arrows show principal runtime flows. The dashed boundary encloses the local application; the dashed oracle link is evaluator-only. C09 selects C07/C08 checks and calls C10; return flows are abbreviated. C08 reads C05 inputs. C11 captures every module. Oracle data never enters retrieval, prompts or human review.

Feasibility. Proposed stack: Python, uv, Typer, Pydantic, SQLite/`Decimal`, PDF extraction and LangGraph checkpoints. FTS5 is the lexical baseline [5]; dense retrieval is an experiment. One local process and a single database writer keep the initial deployment manageable. Typed interfaces enforce evidence provenance and execution permissions.

Agent, reasoning and patterns

2 Adaptive investigation with deterministic calculations

Agentic responsibility. Code establishes provisional facts and arithmetic. The investigator chooses the next unresolved hypothesis, evidence inspection or rule query in response to observations. It cannot change canonical quantities. Each action stores a concise plan, expected evidence and stopping criteria; each result stores an assessment and linked plan revision. These are visible decision records, not private chain-of-thought.



Selected patterns and why they fit

Pattern	ReCycleGuard choice and justification
Tool use	Exact joins, units, quantities, lifecycle checks and allocation conservation remain deterministic C08 operations.
Agentic RAG	C09 chooses queries, assesses evidence grades and re-queries weak results through C07; retrieval is a decision, not a fixed prompt attachment.
Planning / reflection	Persist plans, observations, mistakes and revisions with predecessor links; assess whether the next check adds evidence.
Human review	C13 binds approval, rejection, changes and escalation to exact report and input versions; the agent cannot authenticate a reviewer.
Multi-agent	Optional collector/auditor experiment using the same dispatcher and authorized evidence. It is not a prerequisite for the initial scope.
Durable workflow	LangGraph checkpoints support recovery [6]. Committed application operations and the C11 audit ledger remain authoritative.

Worked adaptation: pending payment is not supported fulfillment

Initial observation: a 100 kg synthetic target has one supported 55 kg allocation, observed twice, and a paid 25 kg item with pending transfer/allocation. Supported fulfillment stays **55 kg**; apparent gap is **45 kg**; **25 kg remains pending**. The two observations do not create 110 kg.

The investigator tests whether the pending item has a completed, eligible allocation. Payment or a newer account snapshot alone does not resolve the finding. Only a later evidenced and reviewed **25 kg allocation** supports a new version with **80 kg** fulfillment and a **20 kg** apparent gap. Without that support, abstain on the unresolved allocation and name the required evidence. A changed dossier starts a new authorized run linked to its parent.

Configurable model gateway and bounded execution

C10 is the sole provider boundary. Select a configuration only after checking structured/schema-constrained responses, typed tool proposals, sufficient context, usage/timing capture and reliability on representative tasks. Pin model/provider, prompt, tool schema and response schema per run. No model is selected here; no automatic fallback is assumed. Optional embeddings use the same gateway and frozen manifests.

Validate every proposal before dispatch. Retain exact permitted requests, responses and errors; secrets stay outside the archive. Permit one budgeted retry per retryable attempt, including invalid structured output if policy allows. Unrecoverable provider failure ends the run; unknown external completion is recorded as `OUTCOME_UNKNOWN`. Resume uses committed operations and consumed budgets, not a fresh allowance. Runtime ceilings are configuration, not measured performance.

Knowledge, memory and storage

3 Preserve originals; retrieve only eligible, versioned evidence

Sources and structure-aware extraction

The planned normative corpus uses official Gazette instruments and CPCB framework/SOP/FAQ/instructions. The 2022 Rules, G.S.R. 801(E), commence on 1 April 2023 [1]; the 8 March 2024 amendment includes Rule 9A [2]. Retain base text and amendments as separate versions rather than declaring one PDF a consolidated rulebook. Clause-specific applicability must be reviewed; a source-class ranking alone cannot decide a legal conflict.

For each source, preserve original bytes, URL, acquisition/capture time, source class, exact instrument/version, commencement, applicable interval, provision path, page/table/cell locator and rights-review decision. Gazette rules, orders, frameworks, SOPs, instructions, FAQs and notice leads remain distinct classes. Public hosting does not automatically authorize model use or redistribution; unknown rights remain `REVIEW_REQUIRED`.

Structure-aware extraction preserves clauses, headings, table rows, headers and footnotes. Every submitted file gets a receipt; every discovered page/row gets a parsed, quarantined, excluded or unavailable disposition. Preserve raw OCR values and validation issues; critical identifiers need deterministic validation or human confirmation. A corrupt file or missing page is not zero business activity.

Data roles. Public aggregate statistics provide context only, never producer transaction evidence. Original synthetic dossiers are the proposed evaluation evidence: manifests, obligations, certificate events, allocations, account snapshots, commercial records and optional PDF/CSV renderings. JSONL records must resolve to declared original evidence; there is no clean answer table secretly available to the investigator.

Retrieval and evidence grading

C06 builds a frozen SQLite FTS5 lexical/BM25 index [5]. C07 filters by access, regime, role and reviewed temporal applicability, including eligible general provisions. Optional dense retrieval is adopted only if measured gains justify it; it uses the same eligibility and provenance contract. No embedding model or vector service is selected.

Expand parent headers, footnotes and applicable amendment counterparts under a recorded traversal policy (default depth 2, at most 20 nodes). Grade **authority, temporal applicability, direct support, completeness and contradiction** separately. Missing signed authority is a lead, not a definitive rule citation. Ambiguous locators or conflicting applicability require abstention/escalation. Preserve ranked candidates, expansion edges, grading, query revisions, not-scored items and context exclusions against full snapshot membership.

Storage and memory: separate meanings, explicit lineage

Layer	Contents and authoritative meaning	Planned storage
Original files	Immutable exact PDF/CSV/JSON/HTML bytes, receipt, hash, rights and access scope.	Content-addressed objects + catalog
Extracted observations	What a specific source page/row says; raw values, revisions and coverage.	JSONL/page text + SQLite links
Events and credit lots	Distinct lifecycle events; supported equivalence links; underlying eligible capacity.	SQLite typed entities / assertions
Supported allocations	Evidenced use of a credit lot against one dated liability bucket; active revisions only.	SQLite allocation ledger
Account comparisons	Captured reported balance, compared with ledger; never an invented allocation.	SQLite observations / deltas
Snapshots and audit	Frozen case/corpus/profile/index membership, full operation payloads, reports and reviews.	Canonical manifests, blobs, runtime SQLite
Hidden benchmark truth	World ledger, corruptions, labels, seeds and visible-evidence expectations; no runtime table.	Separate evaluator-only root

Working memory versus persistent case history

C09 run state and LangGraph checkpoints hold the finding queue, plans, observations, current status and budgets. SQLite plus immutable artifacts retain case history, findings and human decisions. Previous interpretations may aid navigation but are *never automatically reused as legal truth*; each run pins its own reviewed profile and source versions.

Retention is broader than model context. Originals and observations remain available under their access policy even when never retrieved, cited or sent to the model. Checkpoints and indexes are derived; they do not replace evidence or the authoritative audit ledger.

Tools, contracts and governance

4 Typed execution boundaries and accountable human decisions

Shared protocol. Each run freezes its case, corpus, calculation profile, index, code, prompts and policies. Each operation carries actor capability, operation/attempt IDs, idempotency key and audit refs. `ServiceResult<T>` returns status, complete result blob, issues and typed provenance/locator dependencies. Decimal quantities are strings, not binary floats. The representative read/check interfaces below share this protocol.

Tool / owner	Version-bound input	Structured output / explicit failure
`inspect_evidence` (C08)	Artifact, observation or record refs within permitted scope.	Raw/parsed values, provenance, issues; `ACCESS_DENIED`, `MISSING_ARTIFACT`.
`resolve_locator` (C07)	Corpus, source alias/version, legal path, question basis, applicability, expansion policy.	Exact provision revision + evidence bundle; `APPLICABILITY_UNRESOLVED`.
`trace_certificate` (C08)	Certificate/event refs in the pinned case.	Observations, equivalence, event/lot and allocation history; ambiguity stays explicit.
`compute_reconciliation` (C08)	Case/profile, bucket IDs and expected input-membership digest.	Required/supported/pending, gaps and provenance; dimension, precision or profile errors.
`compare_return_draft` (C08)	Draft revision, calculation IDs, section dispositions, deadline refs.	Bucket deltas + scoped readiness; missing/unsupported sections; undetermined applicability.

C08 includes historical allocations and conflicts from the frozen snapshot; model-selected subsets cannot hide liabilities. Unknown/partial data and tool failure remain explicit.

Evidence capabilities and deterministic safeguards

Different documents establish different facts. Payment supports a commercial fact; weight tickets support physical movement; a portal snapshot reports an account position. None alone establishes eligible liability allocation. Recycler EEE capability and compatible metal are separate checks [3,4].

For each metal/liability bucket, required = supplied target + approved signed adjustments; supported = active, evidenced allocations; signed gap = required − supported. Preserve shortfall, surplus and pending separately. Never add metal gaps into a legally meaningful total. Account delta = portal reported − ledger supported; missing portal evidence is null, not zero.

Conservation and identity. Total active use across all events and prior/current buckets cannot exceed supported credit-lot capacity. F02 repeated observations require an authoritative event ID or a complete compound key plus source continuity excluding indistinguishable events; overlap alone is insufficient. F03 legitimate stages sharing a certificate remain distinct events. Check F09 expiry/cancellation. Assess earlier-liability priority *at allocation time*; unknown ordering remains unresolved. Achievement, credit-event and filing cut-offs are separate, source-backed deadline entries [1,2].

Permissions, stopping and report verification

Model-accessible tools read/check and write new derivations, findings or report drafts. Trusted review handlers alone approve profiles/decisions and link supplied acknowledgments. The model cannot mutate source records or portal state, execute arbitrary SQL, shell or dynamic code. Evidence is untrusted input: hostile text is preserved but cannot change tool capabilities, profiles or review status. Redact model inputs through access policy and retain linked permitted originals.

Default limits: **8 investigative tool actions per finding, 2 query reformulations, 1 report repair and 1 retry per retryable attempt**; case-wide action/token/cost/wall-time limits are required configuration. Bound preflight/verification separately. Missing evidence, rule conflicts, no progress or budget exhaustion produce scoped abstention; critical input failure blocks input, invalid unrepaired output blocks output, and unrecoverable provider/audit failure stops execution. C12 rejects legal-compliance verdicts, intent/fraud/concealment allegations and compensation amounts; qualification cannot bypass the prohibition.

Version-bound review and complete audit lineage

Originals → observations → calculations → findings → report → review. C11 retains full tool/model inputs, outputs/errors, contexts, omissions, timing and turn links. Commit domain changes and audit together; audit failure blocks progress. Hash chains detect changes relative to retained anchors.

APPROVE moves `AWAITING_REVIEW` to `COMPLETED`; REJECT or REQUEST_CHANGES to `BLOCKED_OUTPUT`; ESCALATE remains awaiting an external decision. Bind decisions to exact report/dependency hashes. Corrections create a new authorized parent-linked run; approval never transfers. Explicit manual closure records `CLOSED_WITHOUT_DECISION`, never approval. Approved report does not mean filed return.

Evaluation and feasibility

5 Proposed assessment, ownership, limitations and references

Benchmark and controlled comparisons

Plan **48 synthetic dossiers: 24 development and 24 held-out**, split by entity/template/fault combination rather than random rows. Cover all shared families: F01 clean; F02 overlapping export; F03 multistage ID reuse; F04 unit/dimension mix; F05 entity alias; F06 stale snapshot; F07 paid/no transfer; F08 prior arrears; F09 expired/cancelled; F10 capability mismatch; F11 wrong-period rule; F12 unseen notice; F13 weight basis; F14 unsupported regime; F15 extraction loss; F16 injection. This is a pilot target, not an existing or validated dataset.

C15 generates world facts, renders visible evidence and injects faults. Independent reviewed fixtures/calculations establish reference answers; the production calculator is not its own sole oracle. Distinguish hidden world truth from conclusions justified by visible evidence. Investigators get no truth mount, labels, seeds or answer-bearing filenames; C16 scores closed run artifacts, with evaluator views separate from review UI. Test and audit attempted oracle access.

Compare on identical visible inputs and frozen profiles: **(A)** deterministic reconciliation; **(B)** fixed retrieval plus tools; **(C)** bounded single investigator; **(D)** optional collector/auditor. Record actual evidence/call budgets, all failures and repeated trials; freeze configuration before held-out scoring.

Metric	Definition and interpretation
Exact balance agreement	Eligible buckets exactly matching independent decimal references / all eligible buckets. Report mismatches and pending amounts.
Discrepancy precision / recall	Correct discrepancy findings / all predicted; detected reference discrepancies / all reference discrepancies. F01 tests false positives.
Resolution / abstention	Evidence-supported resolutions and correct abstentions over corresponding oracle scopes; do not reward guessing hidden truth.
Citation / temporal correctness	Claim citations that directly support the claim and apply to its role, regime and date / citations assessed, using reviewed references.
Unsupported / prohibited rates	Unsupported claims / assessed claims; prohibited generated verdicts / assessed verdict candidates. Record forbidden tool attempts separately.
Steps, latency and cost	Mean investigative actions per task; measured latency p50/p95 and failures; provider cost per task (unknown cost stays null).
Repeated-run reliability	Repeated trials meeting correctness/safety gates / all trials; report counts, configuration and outcome variation.
Evidence / audit coverage	Accounted assets/pages/rows and fully recorded operations/turns / expected totals; unknown coverage stays unknown.

Hard acceptance gates. Exact supported arithmetic against independent references; complete evidence accounting and operation audit coverage; **zero prohibited generated verdicts in released reports**. Block or qualify unsupported scopes, never invent certainty. Report raw counts alongside rates; unknown coverage is not 100%. Accuracy, latency, cost and collaboration gains remain *unmeasured*. Limits are configured budgets, not benchmark achievements.

Implementation order and team ownership

Build interfaces/audit, ingestion and fixtures, tools, RAG, orchestration, then review/export and evaluation.

Member	Primary responsibility
Abhishek (Lead)	Overall orchestration and correctness, workflow, model gateway and integration.
Sushanth	Data generation and RAG: synthetic dossiers, isolated oracle, indexing and retrieval.
Vedanth	Build and test agent tools for reconciliation, tracing, evidence checks and return comparison.
Sampreeth	Evidence ingestion/storage, audit, human review, reports/export and evaluation integration.

All members review shared interfaces; reference calculations require independent cross-review.

Key limits. Synthetic-only pilot cannot establish production accuracy or legal validity. Source rights, exact EEE codes and reviewed profiles remain activation decisions. Provider/context limits, OCR and dense retrieval require measured selection. Numerical/citation checks cannot prove semantic legal correctness. Local hashes need retained anchors; private evidence requires separate publication/redaction review.

Primary references

[1] MoEFCC. E-Waste (Management) Rules, 2022, G.S.R. 801(E), 2 Nov 2022; Rules 6, 14–15 and Schedules III–IV. Primary source.
[2] MoEFCC. Amendment Rules, 2024, G.S.R. 164(E), 8 Mar 2024; inserted Rule 9A. Primary source.
[3] CPCB. Framework for generation of EPR Certificate; main provisions and Annexure I. Primary source.
[4] CPCB. FAQ under E-Waste Rules, 23 Jan 2024, Section C. Primary source.
[5] SQLite. FTS5 Extension, lexical indexing and BM25. Primary source.
[6] LangChain. LangGraph persistence and checkpoints. Primary source.